

The deep dark web

The unindexed part of the web is also called Deepnet – the invisible, dark or hidden web

256GB flash drive

Kingston launches the 256GB DataTraveler 300 flash drive

Crawling the Deep Web

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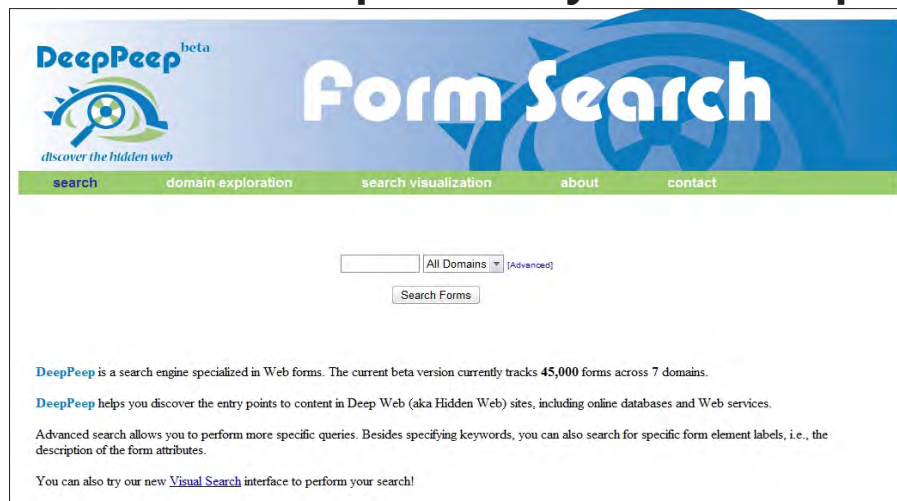
Google has become synonymous with search. Search engines working on algorithms that yield results faster than we can say *search*, makes us believe we have all the information there is. While popular search engines are capable of searching the majority of the web, there are sites that lie below their radar. So there are sites that you probably will never come across. Welcome to *Deep Web*. The web that search engine spiders are able to crawl is called the Surface web, like the iceberg that is visible. On the contrary, the deep web is the huge repository of information that search engine spiders do not have access to.

Why search engines don't search the deep web?

Each search engine has a program called spider, crawler or bot that constantly crawls the internet looking for information. It then indexes all the web pages it crawls through and ranks those pages according to the relevance in content. While crawling web sites, the bots also follow the links on your site thus increasing their footprint. Depending on algorithms, a search engine can either confirm the presence of a page without indexing it, or index the page content and look for hyperlinks on a page. The frequency of the spider crawling web sites depends on the search engine's discretion.

However, search engines have some limitations as they operate on fixed algorithms, often leading to irrelevant results as the search engine is sometimes not able to contextualise the search query. Also search engine bots only crawl static web pages, whereas a majority of the information on the net is stored in databases, which the spiders are not able to crawl. Thus the search results miss out on the data present in databases in universities and government organisations among others. Moreover, there are

A rather large chunk of the web is still undiscovered. We present ways to look deeper



a lot of anonymous surfing web sites that the search engines do not crawl. All this adds up to huge numbers making the search results only a fraction of the total data available.

The deep web

The most logical question right now in your mind will be, if Google can not search the data, what exactly is it and how come it is not crawled. Lets try to decode the deep web content wise.

A database contains information stored in tables which are created by programs such as Access, SQL or Oracle among others. Now this data can only be retrieved by posting a query. The query when executed searches the database to come up with the result that has been specified. This is very different from searching static web pages which can be accessed directly by crawlers.

This database can be text related or non text related like multimedia files, pictures, presentations, PDFs among other things. It may also include dynamically changing content. Many universities, libraries, medical organisations, scientific research organisations have data that cannot appear in popular search engine results. It may only be visible to members who have access rights to those databases.

There are web sites that do not want the search engine crawlers to index their web sites for reasons best know to them.

The dark web

No matter what technology we have, there are some elements in the society that come up with ways to misuse it. There is a sinister side as well to the Deep Web called the Dark Web. A significant portion of the deep web is used by nefarious elements in the society to propagate objectionable material.

Freenet, a distributed, decentralised information storage and retrieval system which allows users to surf the internet anonymously. It has come under a lot of flak as it not only allows users to remain anonymous so that their paths are untraceable, but also hides the fact that someone is using Freenet at all. This has given a lot of leeway to cyber criminals to flourish under the guise of anonymity.

This guise of anonymity though is good for users in countries where internet is highly censored, where freedom of speech is stifled. But at the same time a large number of web sites on paedophilia, terrorist activities, virus coding and other cyber crimes are floating on Freenet.

Another case in point is the Russian mafia arm called Russian Business Net-